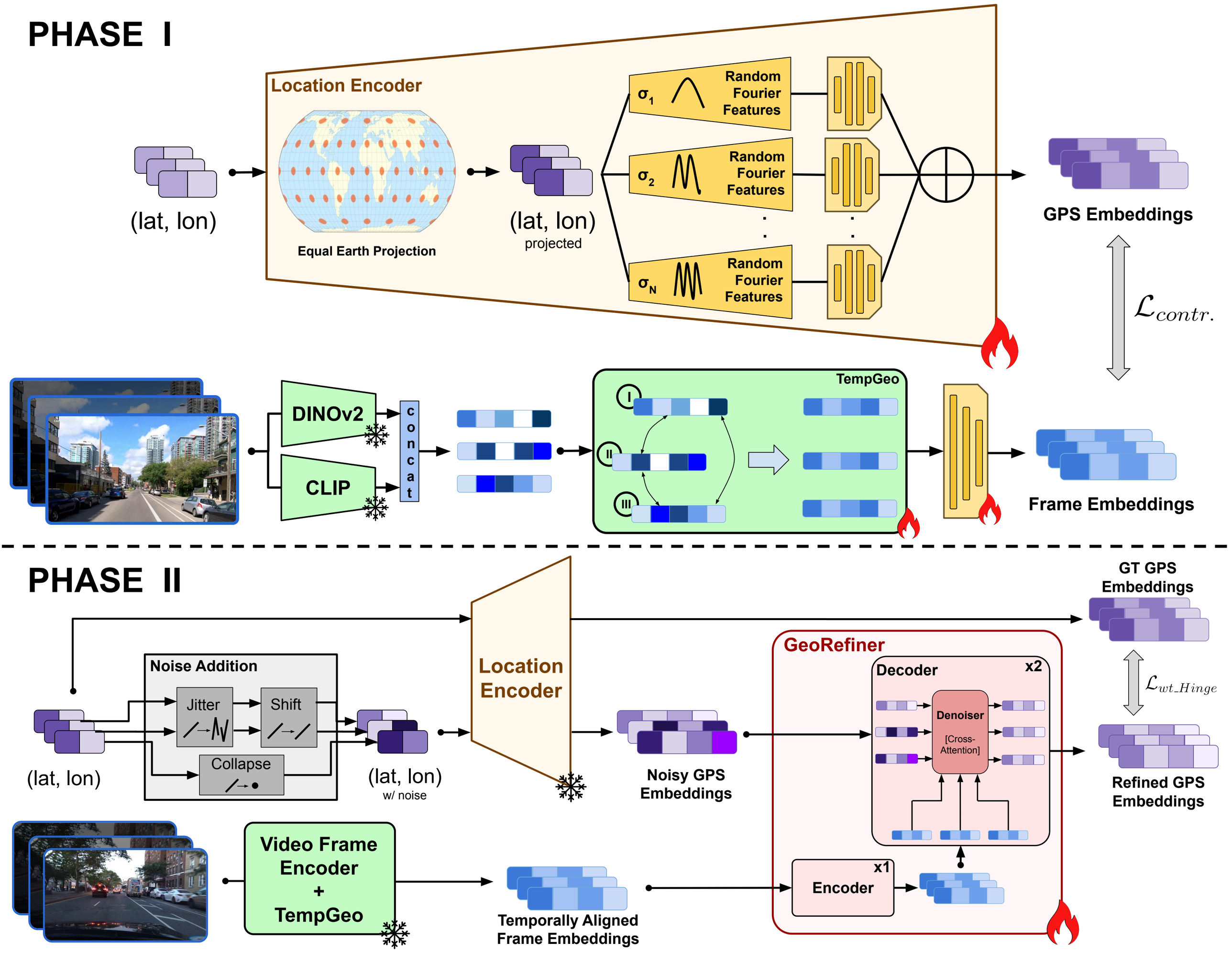
PHASE I
Location Encoder
(lat, lon)
Equal Earth Projection
(lat, lon)
projected
σ1
σ2
σN
Random Fourier Features
GPS Embeddings
$\mathcal{L}_{contr.}$
DINOv2
CLIP
concat
TempGeo
I
II
III
Frame Embeddings
PHASE II
Noise Addition
Jitter
Shift
Collapse
(lat, lon)
(lat, lon)
w/ noise
Location Encoder
Noisy GPS Embeddings
GeoRefiner
Decoder
x2
Denoiser
[Cross-Attention]
Encoder
x1
GT GPS Embeddings
$\mathcal{L}_{wt_Hinge}$
Refined GPS Embeddings
Video Frame Encoder + TempGeo
Temporally Aligned Frame Embeddings